

Causal Survival - Case Base

2026-05-31

Imports

EDA

```
library("Hmisc")  
library("tableone")  
library("xtable")  
library("ggplot2")
```

Cox PH Model

```
library("survival")
```

SMD Balance

```
library("survey")  
library("cobalt")
```

Case Base

```
library("casebase")
```

EDA

Data Set

```
#set seed
seed = 123

#load the dataset
getHdata(rhc)

#re-factor the data set

##main variables of interest

###time: last contact/death date - start date
time <- pmax(rhc$dthdte, rhc$lstctdte, na.rm = TRUE) - rhc$sadmdte

###event: death
Delta <- ifelse(
  rhc$death == "Yes",
  1,
  0
)

###treatment: rhc performed or not
A <- ifelse(
  rhc$swangl == "RHC",
  1,
  0
)

##covriates

###demographics
Gender <- rhc$sex
Age <- as.numeric(rhc$age)
Weight <- as.numeric(rhc$wtkilo1)
Race <- rhc$race
Education <- as.numeric(rhc$edu)
Income <- rhc$income
InsuranceClass <- rhc$ninsclas

###comorbidities
hx_vars <- c(
  "chfhx",
  "cardiohx",
  "dementhx",
  "psychhx",
  "hrpulhx",
  "renalhx",
  "liverhx",
  "gibledhx",
```

```

    "malighx",
    "immunhx",
    "transhx",
    "amihx"
  )

  ###convert labelled data to factors
  rhc[hx_vars] <- lapply(rhc[hx_vars], function(x) {
    x <- as.numeric(x)

    factor(ifelse(x == 1, "Yes", "No"),
            levels = c("No", "Yes"))
  })

  CHFhx <- rhc$chfhx
  CARDIOhx <- rhc$cardiohx
  DEMENThx <- rhc$dementhx
  PSYCHhx <- rhc$psychhx
  CHRPULhx <- rhc$chrpulhx
  RENALhx <- rhc$renalhx
  LIVERhx <- rhc$liverhx
  GIBLEDhx <- rhc$gibledhx
  MALIGHx <- rhc$malighx
  IMMUNhx <- rhc$immunhx
  TRANShx <- rhc$transhx
  AMIhx <- rhc$amihx

  #severities
  Cancer <- rhc$ca
  Category <- rhc$cat1
  Severity <- as.numeric(rhc$das2d3pc)

  #data set design
  data.set <- data.frame(
    time = time,
    Delta = Delta,
    A = A,

    Gender = Gender,
    Age = Age,
    Weight = Weight,
    Race = Race,
    Education = Education,
    Income = Income,
    InsuranceClass = InsuranceClass,

    CHFhx = CHFhx,
    CARDIOhx = CARDIOhx,
    DEMENThx = DEMENThx,
    PSYCHhx = PSYCHhx,
    CHRPULhx = CHRPULhx,
    RENALhx = RENALhx,
    LIVERhx = LIVERhx,

```

```
GIBLEDhx = GIBLEDhx,  
MALIGHx = MALIGHx,  
IMMUNhx = IMMUNhx,  
TRANShx = TRANShx,  
AMIHx = AMIHx,  
  
Cancer = Cancer,  
Category = Category,  
Severity = Severity  
)
```

Table One

```

covariates = c(
  "Gender",
  "Age",
  "Weight",
  "Race",
  "Education",
  "Income",
  "InsuranceClass",

  "CHFhx",
  "CARDIOhx",
  "DEMENThx",
  "PSYCHhx",
  "CHRPULhx",
  "RENALhx",
  "LIVERhx",
  "GIBLEDhx",
  "MALIGHx",
  "IMMUNhx",
  "TRANShx",
  "AMIhx",

  "Cancer",
  "Category",
  "Severity"
)

table.one.original <- CreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set,
  test = FALSE
)
print(table.one.original, smd = TRUE)

```

##		Stratified by A		
##		0	1	SMD
##	n	3551	2184	
##	Gender = Male (%)	1914 (53.9)	1278 (58.5)	0.093
##	Age (mean (SD))	61.76 (17.29)	60.75 (15.63)	0.061
##	Weight (mean (SD))	65.04 (29.50)	72.36 (27.73)	0.256
##	Race (%)			0.036
##	white	2753 (77.5)	1707 (78.2)	
##	black	585 (16.5)	335 (15.3)	
##	asian	0 (0.0)	0 (0.0)	
##	other	213 (6.0)	142 (6.5)	
##	Education (mean (SD))	11.57 (3.13)	11.86 (3.16)	0.091
##	Income (%)			0.142
##	Under \$11k	2081 (58.6)	1145 (52.4)	
##	\$11-\$25k	713 (20.1)	452 (20.7)	
##	\$25-\$50k	500 (14.1)	393 (18.0)	

##	> \$50k	257 (7.2)	194 (8.9)	
##	InsuranceClass (%)			0.194
##	No insurance	186 (5.2)	136 (6.2)	
##	Private	967 (27.2)	731 (33.5)	
##	Medicare	947 (26.7)	511 (23.4)	
##	Medicaid	454 (12.8)	193 (8.8)	
##	Private & Medicare	746 (21.0)	490 (22.4)	
##	Medicare & Medicaid	251 (7.1)	123 (5.6)	
##	CHFhx = Yes (%)	596 (16.8)	425 (19.5)	0.070
##	CARDIOhx = Yes (%)	567 (16.0)	446 (20.4)	0.116
##	DEMENThx = Yes (%)	413 (11.6)	151 (6.9)	0.163
##	PSYCHhx = Yes (%)	286 (8.1)	100 (4.6)	0.143
##	CHRPULhx = Yes (%)	774 (21.8)	315 (14.4)	0.192
##	RENALhx = Yes (%)	149 (4.2)	106 (4.9)	0.032
##	LIVERhx = Yes (%)	265 (7.5)	136 (6.2)	0.049
##	GIBLEDhx = Yes (%)	131 (3.7)	54 (2.5)	0.070
##	MALIGHx = Yes (%)	872 (24.6)	444 (20.3)	0.101
##	IMMUNhx = Yes (%)	907 (25.5)	636 (29.1)	0.080
##	TRANShx = Yes (%)	335 (9.4)	327 (15.0)	0.170
##	AMIHx = Yes (%)	105 (3.0)	95 (4.3)	0.074
##	Cancer (%)			0.107
##	No	2652 (74.7)	1727 (79.1)	
##	Yes	638 (18.0)	334 (15.3)	
##	Metastatic	261 (7.4)	123 (5.6)	
##	Category (%)			0.583
##	ARF	1581 (44.5)	909 (41.6)	
##	COPD	399 (11.2)	58 (2.7)	
##	CHF	247 (7.0)	209 (9.6)	
##	Cirrhosis	175 (4.9)	49 (2.2)	
##	Coma	341 (9.6)	95 (4.3)	
##	Colon Cancer	6 (0.2)	1 (0.0)	
##	Lung Cancer	34 (1.0)	5 (0.2)	
##	MOSF w/Malignancy	241 (6.8)	158 (7.2)	
##	MOSF w/Sepsis	527 (14.8)	700 (32.1)	
##	Missing	0 (0.0)	0 (0.0)	
##	Severity (mean (SD))	20.37 (5.48)	20.70 (5.03)	0.063

Propensity Scores

```
#set seed for output reproducibility
set.seed(seed)

#generate propensity scores
propensity.score <- fitted(
  glm(
    A ~ Gender +
      Age +
      Weight +
      Race +
      Education +
      Income +
      InsuranceClass +
      CHFhx +
      CARDIOhx +
      DEMENThx +
      PSYCHhx +
      CHRPULhx +
      RENALhx +
      LIVERhx +
      GIBLEDhx +
      MALIGHx +
      IMMUNhx +
      TRANShx +
      AMIhx +
      Cancer +
      Category +
      Severity,
    data = data.set,
    family = binomial(link = "logit")
  )
)

#separate propensity scores for exposed & unexposed groups
propensity.score.0 <- propensity.score[A == 0]
propensity.score.1 <- propensity.score[A == 1]

#visualize the propensity scores overlap
par(
  mar=c(2,2,2,0)
)

hist(
  propensity.score.0,
  col=rgb(0,0,1,0.5),
  breaks=seq(0,1,by=0.02),
  ylim=c(0,350),
  main="Propensity Score Overlap",
  xlab="Propensity Scores"
)
```

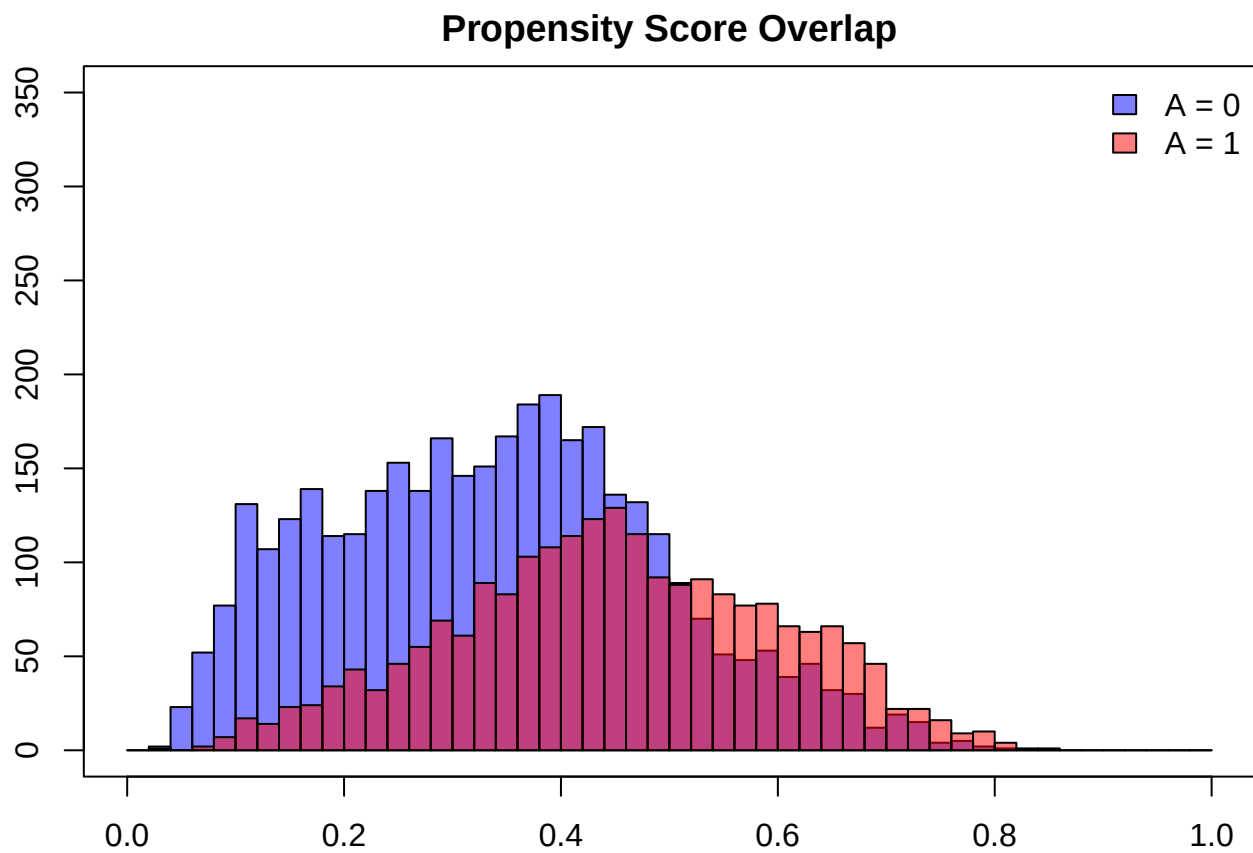
```

hist(
  propensity.score.1,
  col=rgb(1,0,0,0.5),
  breaks=seq(0,1,by=0.02),
  add=T
)

box()

legend(
  "topright",
  legend=c('A = 0', 'A = 1'),
  fill=c(rgb(0,0,1,0.5),rgb(1,0,0,0.5)),
  border="black", bty="n"
)

```



Stabilized IPW

```
p.A1 <- mean(A)

#calculate unstabilized ipws
ipw.unstabilized <- ifelse(
  A == 1,
  1 / propensity.score,
  1 / (1 - propensity.score)
)

#calculate stabilized ipws
ipw.stabilized <- ifelse(
  A == 1,
  p.A1 / propensity.score,
  (1 - p.A1) / (1 - propensity.score)
)

#visualize the ipws
par(
  mar=c(2,2,2,0)
)

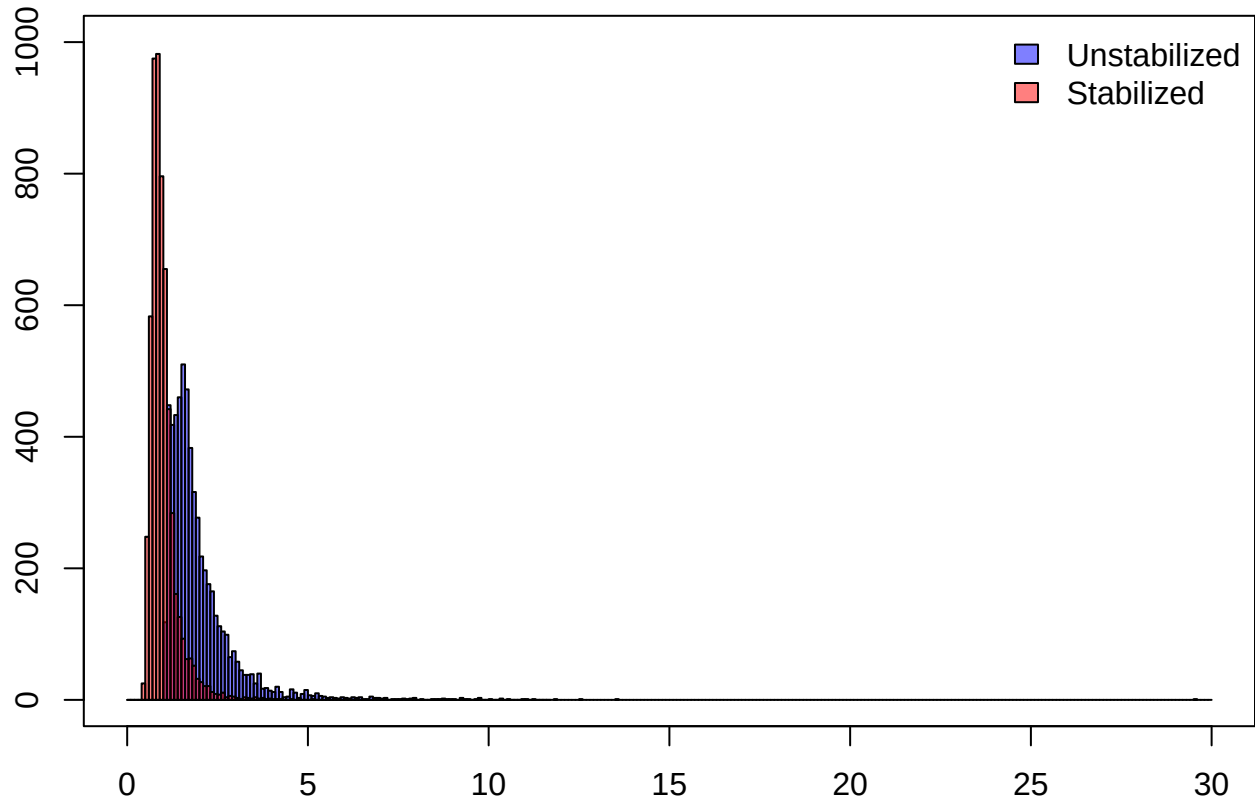
hist(
  ipw.unstabilized,
  col=rgb(0,0,1,0.5),
  breaks=seq(0, max(ipw.unstabilized) + 0.5,by=0.1),
  ylim=c(0,1000),
  main="Unstabilized Vs. Stabilized IPWs",
  xlab="IPWs"
)

hist(
  ipw.stabilized,
  col=rgb(1,0,0,0.5),
  breaks=seq(0, max(ipw.stabilized) + 0.5,by=0.1),
  add=T
)

box()

legend(
  "topright",
  legend=c('Unstabilized', 'Stabilized'),
  fill=c(rgb(0,0,1,0.5),rgb(1,0,0,0.5)),
  border="black", bty="n"
)
```

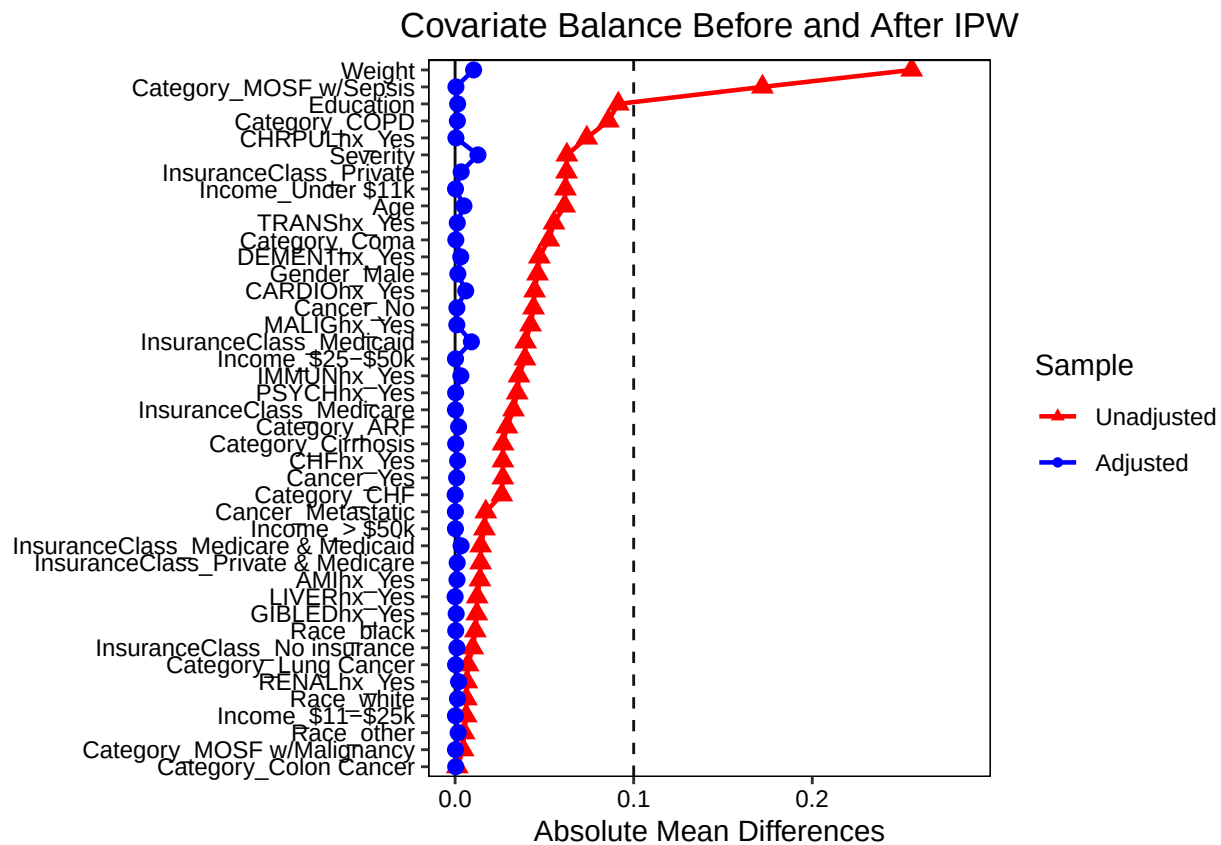
Unstablized Vs. Stabilized IPWs



Covariate Balance

```
#calculate the covariate balance before & after IPW
balance <- bal.tab(
  A ~ Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  weights = ipw.stabilized,
  method = "weighting",
  estimand = "Average Treatment Effect",
  un = TRUE
)

#visualize the covariate balance before & after IPW
love.plot(
  balance,
  stats = "mean.diffs",
  threshold = 0.1,
  abs = TRUE,
  line = TRUE,
  var.order = "unadjusted",
  colors = c("red", "blue"),
  shapes = c(17, 16),
  limits = c(m = c(0, 0.3)),
  title = "Covariate Balance Before and After IPW",
  labels = TRUE
)
```



Note that covariate balance improved substantially after the inverse probability weighting (IPW) procedure was applied to create the pseudo-population.

Table Two

```
data.set.ipw <- svydesign(
  ids = ~ 0,
  data = data.set,
  weights = ipw.stabilized
)

table.one.ipw <- svyCreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set.ipw,
  test = FALSE
)

print(table.one.ipw, smd = TRUE)
```

##	Stratified by A		
##	0	1	SMD
## n	3554.6	2178.4	
## Gender = Male (%)	1968.2 (55.4)	1209.5 (55.5)	0.003
## Age (mean (SD))	61.33 (17.24)	61.25 (15.74)	0.005
## Weight (mean (SD))	67.93 (29.11)	68.23 (28.44)	0.010
## Race (%)			0.007
## white	2759.7 (77.6)	1694.1 (77.8)	
## black	576.6 (16.2)	354.1 (16.3)	
## asian	0.0 (0.0)	0.0 (0.0)	
## other	218.3 (6.1)	130.1 (6.0)	
## Education (mean (SD))	11.69 (3.14)	11.69 (3.08)	0.001
## Income (%)			0.001
## Under \$11k	1997.5 (56.2)	1224.6 (56.2)	
## \$11-\$25k	723.4 (20.4)	443.7 (20.4)	
## \$25-\$50k	555.6 (15.6)	340.0 (15.6)	
## > \$50k	278.1 (7.8)	170.0 (7.8)	
## InsuranceClass (%)			0.031
## No insurance	198.0 (5.6)	119.1 (5.5)	
## Private	1047.8 (29.5)	634.7 (29.1)	
## Medicare	899.1 (25.3)	550.6 (25.3)	
## Medicaid	404.3 (11.4)	267.6 (12.3)	
## Private & Medicare	775.2 (21.8)	472.6 (21.7)	
## Medicare & Medicaid	230.1 (6.5)	133.7 (6.1)	
## CHFhx = Yes (%)	634.9 (17.9)	392.2 (18.0)	0.004
## CARDIOhx = Yes (%)	636.8 (17.9)	403.2 (18.5)	0.015
## DEMENThx = Yes (%)	347.5 (9.8)	206.1 (9.5)	0.011
## PSYCHhx = Yes (%)	237.3 (6.7)	146.1 (6.7)	0.001
## CHRPULhx = Yes (%)	669.0 (18.8)	411.0 (18.9)	0.001
## RENALhx = Yes (%)	161.7 (4.5)	103.3 (4.7)	0.009
## LIVERhx = Yes (%)	250.2 (7.0)	153.3 (7.0)	<0.001
## GIBLEDhx = Yes (%)	114.7 (3.2)	69.0 (3.2)	0.003
## MALIGHx = Yes (%)	817.4 (23.0)	503.0 (23.1)	0.002
## IMMUNhx = Yes (%)	965.2 (27.2)	598.5 (27.5)	0.007
## TRANShx = Yes (%)	412.5 (11.6)	255.5 (11.7)	0.004
## AMIhx = Yes (%)	122.4 (3.4)	77.3 (3.5)	0.006

##	Cancer (%)			0.002
##	No	2712.1 (76.3)	1659.9 (76.2)	
##	Yes	603.9 (17.0)	371.9 (17.1)	
##	Metastatic	238.5 (6.7)	146.5 (6.7)	
##	Category (%)			0.011
##	ARF	1541.0 (43.4)	948.6 (43.5)	
##	COPD	282.7 (8.0)	170.5 (7.8)	
##	CHF	282.7 (8.0)	173.3 (8.0)	
##	Cirrhosis	138.6 (3.9)	85.6 (3.9)	
##	Coma	270.2 (7.6)	164.7 (7.6)	
##	Colon Cancer	4.4 (0.1)	3.4 (0.2)	
##	Lung Cancer	24.1 (0.7)	14.2 (0.7)	
##	MOSF w/Malignancy	246.5 (6.9)	150.6 (6.9)	
##	MOSF w/Sepsis	764.4 (21.5)	467.4 (21.5)	
##	Missing	0.0 (0.0)	0.0 (0.0)	
##	Severity (mean (SD))	20.48 (5.53)	20.55 (4.95)	0.013

Estimation

Assumptions

Non-Informative Censoring

Model 1: Unadjusted Cox

```
#set seed for output reproducibility
set.seed(seed)
```

```
#fit model
model.1 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  robust = TRUE
)
```

```
#generate summary
summary(model.1)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, robust = TRUE)
##
##      n= 5735, number of events= 3722
##
##      coef exp(coef) se(coef) robust se      z Pr(>|z|)
## A 0.15198  1.16414  0.03351  0.03420 4.444 8.85e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## A      1.164      0.859      1.089      1.245
##
## Concordance= 0.527 (se = 0.005 )
## Likelihood ratio test= 20.35 on 1 df,  p=6e-06
## Wald test               = 19.75 on 1 df,  p=9e-06
## Score (logrank) test = 20.62 on 1 df,  p=6e-06,  Robust = 19.74  p=9e-06
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Model 2: Unadjusted Case Base

```
#set seed for output reproducibility
set.seed(seed)
```

```
#fit model
model.2 <- fitSmoothHazard(
  Delta ~ time * A,
  data = data.set,
  ratio = 10
)
```

```
## 'time' will be used as the time variable
```

```
#generate summary
summary(model.2)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 37220
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ time * A, data = data.set,
##   ratio = 10)
##
## Coefficients:
##           Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -5.4427656  0.0292176 -186.284  < 2e-16 ***
## time        -0.0013353  0.0001062  -12.579  < 2e-16 ***
## A            0.3276038  0.0462950   7.076 1.48e-12 ***
## time:A       -0.0009249  0.0001877  -4.927 8.37e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 24945  on 40941  degrees of freedom
## Residual deviance: 24403  on 40938  degrees of freedom
## AIC: 24411
##
## Number of Fisher Scoring iterations: 6
```

```
#HR
cat("HR-hat:", exp(-model.2$coefficients[3]))
```

```
## HR-hat: 0.7206485
```

Model 3: Covariate Adjusted Cox

```
#set seed for output reproducibility
set.seed(seed)
```

```
#fit model
model.3 <- coxph(
  Surv(time, Delta) ~ A +
    Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  robust = TRUE
)
```

```
#generate summary
summary(model.3)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A + Gender + Age + Weight +
##       Race + Education + Income + InsuranceClass + CHFhx + CARDIOhx +
##       DEMENThx + PSYCHhx + CHRPULhx + RENALhx + LIVERhx + GIBLEDhx +
##       MALIGHx + IMMUNhx + TRANShx + AMIhx + Cancer + Category +
##       Severity, data = data.set, robust = TRUE)
```

```
## n= 5735, number of events= 3722
```

	coef	exp(coef)	se(coef)	robust se
## A	0.2533449	1.2883275	0.0357613	0.0376554
## GenderMale	0.1233932	1.1313292	0.0345960	0.0361201
## Age	0.0106143	1.0106709	0.0014473	0.0015407
## Weight	-0.0020044	0.9979976	0.0006072	0.0006244
## Raceblack	0.0205835	1.0207968	0.0480552	0.0491244
## Raceasian	NA	NA	0.0000000	0.0000000

## Raceother	0.1232340	1.1311490	0.0707914	0.0682031
## Education	0.0054062	1.0054208	0.0060673	0.0062343
## Income\$11-\$25k	-0.1325086	0.8758954	0.0458674	0.0467457
## Income\$25-\$50k	-0.2138434	0.8074749	0.0562912	0.0575050
## Income> \$50k	-0.1785389	0.8364916	0.0744784	0.0769455
## InsuranceClassPrivate	-0.0360729	0.9645700	0.0834823	0.0830787
## InsuranceClassMedicare	-0.0569048	0.9446840	0.0854888	0.0853701
## InsuranceClassMedicaid	-0.1679833	0.8453679	0.0931216	0.0941160
## InsuranceClassPrivate & Medicare	-0.0871831	0.9165093	0.0873752	0.0877396
## InsuranceClassMedicare & Medicaid	-0.1063721	0.8990900	0.0996189	0.0989352
## CHFhxYes	0.0915364	1.0958567	0.0505137	0.0505770
## CARDIOhxYes	-0.0508656	0.9504064	0.0480933	0.0533188
## DEMENThxYes	0.1334706	1.1427877	0.0544424	0.0595816
## PSYCHhxYes	-0.1652208	0.8477065	0.0704540	0.0691374
## CHRPULhxYes	0.0935860	1.0981051	0.0504644	0.0489547
## RENALhxYes	0.1684734	1.1834968	0.0790452	0.0854669
## LIVERhxYes	0.6040056	1.8294320	0.0941283	0.1033602
## GIBLEDhxYes	-0.0280961	0.9722950	0.1159686	0.1252369
## MALIGHxYes	-0.0408440	0.9599789	0.1780595	0.1854789
## IMMUNhxYes	0.0621781	1.0641519	0.0382973	0.0399256
## TRANShxYes	0.0220129	1.0222570	0.0559133	0.0567055
## AMIhxYes	-0.0135365	0.9865547	0.0917747	0.0843918
## CancerYes	0.3383995	1.4027007	0.1801795	0.1875629
## CancerMetastatic	0.5557491	1.7432464	0.2045181	0.2086943
## CategoryCOPD	-0.2602465	0.7708615	0.0775658	0.0696279
## CategoryCHF	-0.3735585	0.6882807	0.0787545	0.0751274
## CategoryCirrhosis	0.0548676	1.0564007	0.1195645	0.1318199
## CategoryComa	1.0743080	2.9279660	0.0617195	0.0740730
## CategoryColon Cancer	-0.3422784	0.7101504	0.5064276	0.3328934
## CategoryLung Cancer	0.2756386	1.3173716	0.1831278	0.1423742
## CategoryMOSF w/Malignancy	0.6104038	1.8411746	0.0748845	0.0775313
## CategoryMOSF w/Sepsis	0.1153416	1.1222568	0.0451955	0.0480631
## CategoryMissing	NA	NA	0.0000000	0.0000000
## Severity	-0.0297207	0.9707166	0.0034800	0.0033148
##	z Pr(> z)			
## A	6.728	1.72e-11	***	
## GenderMale	3.416	0.000635	***	
## Age	6.889	5.60e-12	***	
## Weight	-3.210	0.001327	**	
## Raceblack	0.419	0.675211		
## Raceasian	NA	NA		
## Raceother	1.807	0.070783	.	
## Education	0.867	0.385851		
## Income\$11-\$25k	-2.835	0.004587	**	
## Income\$25-\$50k	-3.719	0.000200	***	
## Income> \$50k	-2.320	0.020323	*	
## InsuranceClassPrivate	-0.434	0.664143		
## InsuranceClassMedicare	-0.667	0.505049		
## InsuranceClassMedicaid	-1.785	0.074285	.	
## InsuranceClassPrivate & Medicare	-0.994	0.320390		
## InsuranceClassMedicare & Medicaid	-1.075	0.282299		
## CHFhxYes	1.810	0.070320	.	
## CARDIOhxYes	-0.954	0.340088		
## DEMENThxYes	2.240	0.025082	*	

```

## PSYCHhxYes          -2.390 0.016860 *
## CHRPUHhxYes         1.912 0.055916 .
## RENALhxYes          1.971 0.048699 *
## LIVERhxYes          5.844 5.11e-09 ***
## GIBLEDhxYes         -0.224 0.822490
## MALIGHhxYes         -0.220 0.825709
## IMMUNhxYes          1.557 0.119388
## TRANShxYes          0.388 0.697870
## AMIhxYes            -0.160 0.872566
## CancerYes           1.804 0.071201 .
## CancerMetastatic    2.663 0.007745 **
## CategoryCOPD        -3.738 0.000186 ***
## CategoryCHF         -4.972 6.62e-07 ***
## CategoryCirrhosis   0.416 0.677241
## CategoryComa        14.503 < 2e-16 ***
## CategoryColon Cancer -1.028 0.303859
## CategoryLung Cancer  1.936 0.052866 .
## CategoryMOSF w/Malignancy 7.873 3.46e-15 ***
## CategoryMOSF w/Sepsis 2.400 0.016404 *
## CategoryMissing      NA      NA
## Severity            -8.966 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
##               exp(coef) exp(-coef) lower .95 upper .95
## A               1.2883    0.7762    1.1967    1.3870
## GenderMale      1.1313    0.8839    1.0540    1.2143
## Age             1.0107    0.9894    1.0076    1.0137
## Weight          0.9980    1.0020    0.9968    0.9992
## Raceblack       1.0208    0.9796    0.9271    1.1240
## Raceasian       NA        NA        NA        NA
## Raceother       1.1311    0.8841    0.9896    1.2929
## Education       1.0054    0.9946    0.9932    1.0178
## Income$11-$25k  0.8759    1.1417    0.7992    0.9599
## Income$25-$50k  0.8075    1.2384    0.7214    0.9038
## Income> $50k    0.8365    1.1955    0.7194    0.9727
## InsuranceClassPrivate 0.9646    1.0367    0.8196    1.1351
## InsuranceClassMedicare 0.9447    1.0586    0.7991    1.1167
## InsuranceClassMedicaid 0.8454    1.1829    0.7030    1.0166
## InsuranceClassPrivate & Medicare 0.9165    1.0911    0.7717    1.0885
## InsuranceClassMedicare & Medicaid 0.8991    1.1122    0.7406    1.0915
## CHFhxYes        1.0959    0.9125    0.9924    1.2101
## CARDIOhxYes     0.9504    1.0522    0.8561    1.0551
## DEMENThxYes     1.1428    0.8751    1.0168    1.2843
## PSYCHhxYes      0.8477    1.1797    0.7403    0.9707
## CHRPUHhxYes     1.0981    0.9107    0.9976    1.2087
## RENALhxYes      1.1835    0.8450    1.0010    1.3993
## LIVERhxYes      1.8294    0.5466    1.4939    2.2403
## GIBLEDhxYes     0.9723    1.0285    0.7607    1.2428
## MALIGHhxYes     0.9600    1.0417    0.6674    1.3808
## IMMUNhxYes      1.0642    0.9397    0.9841    1.1508
## TRANShxYes      1.0223    0.9782    0.9147    1.1424
## AMIhxYes        0.9866    1.0136    0.8362    1.1640
## CancerYes       1.4027    0.7129    0.9712    2.0259

```

## CancerMetastatic	1.7432	0.5736	1.1580	2.6242
## CategoryCOPD	0.7709	1.2972	0.6725	0.8836
## CategoryCHF	0.6883	1.4529	0.5940	0.7975
## CategoryCirrhosis	1.0564	0.9466	0.8159	1.3678
## CategoryComa	2.9280	0.3415	2.5323	3.3855
## CategoryColon Cancer	0.7102	1.4082	0.3698	1.3637
## CategoryLung Cancer	1.3174	0.7591	0.9966	1.7414
## CategoryMOSF w/Malignancy	1.8412	0.5431	1.5816	2.1433
## CategoryMOSF w/Sepsis	1.1223	0.8911	1.0214	1.2331
## CategoryMissing	NA	NA	NA	NA
## Severity	0.9707	1.0302	0.9644	0.9770

##

Concordance= 0.663 (se = 0.005)

Likelihood ratio test= 893.8 on 38 df, p=<2e-16

Wald test = 880.2 on 38 df, p=<2e-16

Score (logrank) test = 982.3 on 38 df, p=<2e-16, Robust = 862.6 p=<2e-16

##

(Note: the likelihood ratio and score tests assume independence of

observations within a cluster, the Wald and robust score tests do not).

Model 4: Covariate Adjusted Case Base

```
#set seed for output reproducibility
set.seed(seed)
```

```
#fit model
model.4 <- fitSmoothHazard(
  Delta ~ time * A +
  Gender +
  Age +
  Weight +
  Race +
  Education +
  Income +
  InsuranceClass +
  CHFhx +
  CARDIOhx +
  DEMENThx +
  PSYCHhx +
  CHRPULhx +
  RENALhx +
  LIVERhx +
  GIBLEDhx +
  MALIGHx +
  IMMUNhx +
  TRANShx +
  AMIhx +
  Cancer +
  Category +
  Severity,
  data = data.set,
  ratio = 10
)
```

```
## 'time' will be used as the time variable
```

```
#generate summary
summary(model.4)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 37220
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ time * A + Gender + Age + Weight +
##   Race + Education + Income + InsuranceClass + CHFhx + CARDIOhx +
##   DEMENThx + PSYCHhx + CHRPULhx + RENALhx + LIVERhx + GIBLEDhx +
##   MALIGHx + IMMUNhx + TRANShx + AMIhx + Cancer + Category +
##   Severity, data = data.set, ratio = 10)
```

```

##
## Coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -5.7116321  0.1760828 -32.437 < 2e-16 ***
## time          -0.0011700  0.0001049 -11.152 < 2e-16 ***
## A              0.4426609  0.0486895  9.092 < 2e-16 ***
## GenderMale     0.1394798  0.0371156  3.758 0.000171 ***
## Age            0.0120684  0.0015366  7.854 4.03e-15 ***
## Weight        -0.0017427  0.0006489 -2.686 0.007241 **
## Raceblack      0.0356856  0.0514970  0.693 0.488331
## Raceother      0.1811505  0.0759828  2.384 0.017121 *
## Education      0.0021975  0.0065839  0.334 0.738554
## Income$11-$25k -0.1241322  0.0490603 -2.530 0.011400 *
## Income$25-$50k -0.2535795  0.0602253 -4.211 2.55e-05 ***
## Income> $50k   -0.1269570  0.0806755 -1.574 0.115563
## InsuranceClassPrivate -0.0484791  0.0892232 -0.543 0.586891
## InsuranceClassMedicare -0.0335931  0.0910923 -0.369 0.712291
## InsuranceClassMedicaid -0.1866507  0.0986203 -1.893 0.058409 .
## InsuranceClassPrivate & Medicare -0.0860499  0.0933530 -0.922 0.356649
## InsuranceClassMedicare & Medicaid -0.1595823  0.1059975 -1.506 0.132188
## CHFhxYes       0.1065191  0.0544413  1.957 0.050396 .
## CARDIOhxYes    -0.0214700  0.0518465 -0.414 0.678795
## DEMENThxYes     0.1452747  0.0589155  2.466 0.013670 *
## PSYCHhxYes     -0.1532728  0.0738139 -2.076 0.037850 *
## CHRPUHhxYes     0.1124540  0.0545361  2.062 0.039207 *
## RENALhxYes      0.1502453  0.0846076  1.776 0.075768 .
## LIVERhxYes      0.6496503  0.1008321  6.443 1.17e-10 ***
## GIBLEDhxYes    -0.1017952  0.1245216 -0.817 0.413648
## MALIGHhxYes    -0.0683466  0.1999252 -0.342 0.732456
## IMMUNhxYes      0.0741173  0.0412018  1.799 0.072037 .
## TRANShxYes      0.0352740  0.0599983  0.588 0.556588
## AMIhxYes       -0.0688724  0.0977449 -0.705 0.481051
## CancerYes       0.4087569  0.2019146  2.024 0.042929 *
## CancerMetastatic 0.6260354  0.2278077  2.748 0.005994 **
## CategoryCOPD    -0.3549760  0.0817852 -4.340 1.42e-05 ***
## CategoryCHF     -0.4600254  0.0830521 -5.539 3.04e-08 ***
## CategoryCirrhosis 0.1148128  0.1290092  0.890 0.373488
## CategoryComa     1.2560536  0.0701372 17.909 < 2e-16 ***
## CategoryColon Cancer 0.1083875  0.5374134  0.202 0.840164
## CategoryLung Cancer 0.2502600  0.2000100  1.251 0.210848
## CategoryMOSF w/Malignancy 0.7787844  0.0828317  9.402 < 2e-16 ***
## CategoryMOSF w/Sepsis 0.1368959  0.0480158  2.851 0.004357 **
## Severity        -0.0351639  0.0036987 -9.507 < 2e-16 ***
## time:A          -0.0009303  0.0001846 -5.040 4.66e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 24945  on 40941  degrees of freedom
## Residual deviance: 23382  on 40901  degrees of freedom
## AIC: 23464
##
## Number of Fisher Scoring iterations: 6

```

```
#HR  
cat("HR-hat:", exp(-model.4$coefficients[3]))
```

```
## HR-hat: 0.642325
```

Model 5: Marginal Structural Cox

```
#set seed for output reproducibility
set.seed(seed)
```

```
#fit model
model.5 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  weights = ipw.stabilized,
  robust = TRUE
)
```

```
#generate summary
summary(model.5)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, weights = ipw.stabilized,
##       robust = TRUE)
##
##      n= 5735, number of events= 3722
##
##      coef exp(coef) se(coef) robust se      z Pr(>|z|)
## A 0.20890   1.23232  0.03338   0.03764 5.55 2.86e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## A      1.232      0.8115      1.145      1.327
##
## Concordance= 0.535 (se = 0.005 )
## Likelihood ratio test= 38.61 on 1 df,  p=5e-10
## Wald test               = 30.8 on 1 df,  p=3e-08
## Score (logrank) test = 39.32 on 1 df,  p=4e-10, Robust = 30.69 p=3e-08
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Model 6: IPW Case Base PH

```
#set seed for output reproducibility
set.seed(seed)

#fit model
model.6 <- fitSmoothHazard(
  Delta ~ time * A,
  data = data.set,
  ratio = 10,
  family = "glm",
  weights = ipw.stabilized
)

#generate summary
summary(model.6)

## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 37220
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ time * A, data = data.set,
##   family = "glm", ratio = 10, weights = ipw.stabilized)
##
## Coefficients:
##               Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -5.4427656  0.0292176 -186.284  < 2e-16 ***
## time        -0.0013353  0.0001062  -12.579  < 2e-16 ***
## A            0.3276038  0.0462950   7.076 1.48e-12 ***
## time:A       -0.0009249  0.0001877  -4.927 8.37e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##   Null deviance: 24945  on 40941  degrees of freedom
## Residual deviance: 24403  on 40938  degrees of freedom
## AIC: 24411
##
## Number of Fisher Scoring iterations: 6

#HR
cat("HR-hat:", exp(-model.6$coefficients[3]))

## HR-hat: 0.7206485
```

Model 7: IPW Case Base (Variable Hazard over Time)

```
#set seed for output reproducibility
set.seed(seed)
```

```
#fit model
model.7 <- fitSmoothHazard(
  Delta ~ pspline(time, df = 2) * A,
  data = data.set,
  ratio = 10,
  weights = ipw.stabilized
)
```

```
#generate summary
summary(model.7)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 5735
## Number of events: 3722
## Number of base moments: 37220
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ pspline(time, df = 2) * A,
##   data = data.set, ratio = 10, weights = ipw.stabilized)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      23.6590      1.0111  23.400  <2e-16 ***
## pspline(time, df = 2)1  -34.4883      1.2556 -27.467  <2e-16 ***
## pspline(time, df = 2)2  -28.1435      0.8834 -31.858  <2e-16 ***
## pspline(time, df = 2)3  -31.1686      1.1479 -27.153  <2e-16 ***
## pspline(time, df = 2)4  -28.3014      0.9687 -29.216  <2e-16 ***
## pspline(time, df = 2)5  -30.1687      1.2920 -23.351  <2e-16 ***
## pspline(time, df = 2)6  -26.5612      2.1541 -12.331  <2e-16 ***
## pspline(time, df = 2)7  -23.0767     24.7448  -0.933   0.3510
## A                   3.3887      1.6687   2.031   0.0423 *
## pspline(time, df = 2)1:A  -3.7923      2.0787  -1.824   0.0681 .
## pspline(time, df = 2)2:A  -3.2169      1.4510  -2.217   0.0266 *
## pspline(time, df = 2)3:A  -3.7191      1.9093  -1.948   0.0514 .
## pspline(time, df = 2)4:A  -4.0074      1.6010  -2.503   0.0123 *
## pspline(time, df = 2)5:A  -2.9202      2.0900  -1.397   0.1624
## pspline(time, df = 2)6:A  -6.2196      3.1995  -1.944   0.0519 .
## pspline(time, df = 2)7:A   1.4247     28.4723   0.050   0.9601
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 24945  on 40941  degrees of freedom
## Residual deviance: 21766  on 40926  degrees of freedom
## AIC: 21798
```

```
##  
## Number of Fisher Scoring iterations: 6
```

```
#plot hazard-time  
plot.time <- seq(  
  min(data.set$time, na.rm = TRUE),  
  max(data.set$time, na.rm = TRUE),  
  by = 0.1  
)  
  
plot.data <- data.frame(  
  A = factor(  
    "No RHC",  
    levels = c("No RHC", "RHC")  
  ),  
  time = plot.time  
)  
  
plot(  
  model.7,  
  type = "hr",  
  newdata = plot.data,  
  var = "A",  
  xvar = "time",  
  ci = TRUE,  
  ylim = c(0,2)  
)
```

